S. MOHAMMAD HEJAZI

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MASc candidate in EECS working on the reliability, reproducibility, and cost of LLM-based software engineering agents. Experience evaluating coding agents on SWE-bench and building reproducibility tooling for LLM systems.

EXPERIENCE

Grimlock Consulting Sep 2026 – Present
Research Intern (part-time) *Remote*

- Run experiments to improve AI coding agents (e.g., Claude Code), testing techniques that raise resolve rates and cut cost on SWE-bench Verified, SWE-bench Pro, and related benchmarks.
- Analyze agent trajectories and logs; deliver weekly reports with charts and metrics to guide next steps.

York University Jan 2025 – Present
Graduate Research Assistant, supervised by Prof. Zhen Ming (Jack) Jiang *Toronto, Canada*

- Built a reproducibility harness for LLM-powered systems that combines execution stabilization with record/replay, achieving exact replay across 4 workloads, 6 open-weight LLMs, vLLM and llama.cpp, and A100/H100/H200 GPUs. Default runs reproduced only 58–64% of SWE-agent trajectories exactly.
- Identified 11 sources of non-determinism across the LLM execution stack, 3 of them previously undocumented.
- Developing a process-aware framework to assess benchmark consistency of LLM-powered SE agents, using trajectories and divergence-guided replay instead of exhaustive repeated runs (early stage).
- Exploratory study of cost and failure modes of SWE-bench Verified agents (SWE-agent, mini-SWE-agent, Moatless Tools, CORTEXA) from official logs; prototyped an early-stopping technique to cut wasted cost.

York University 2025 – 2026
Teaching Assistant *Toronto, Canada*

- Data Structures, Algorithms, Advanced OOP: grading, labs, tutorials. Earlier TA at Amirkabir (2022).

Qom Province Electricity Distribution Company Jun 2021 – Sep 2021
Intern, Smart Power Grid *Qom, Iran*

- Surveyed ML for power distribution (forecasting, anomaly detection); authored a smart-grid survey.

EDUCATION

York University, Toronto, Canada *Jan 2025 – Expected Apr 2027*
Master of Applied Science, Electrical Engineering & Computer Science GPA: 8.25/9.00

Amirkabir University of Technology, Tehran, Iran *Sept 2018 – Jul 2023*
Bachelor of Science, Computer Engineering GPA: 18.24/20 (3.89/4)

Honors: University Honor Student (2023); top 0.8% in Iran's national university entrance exam (2018).

PUBLICATIONS

– **S. Mohammad Hejazi** and Z. M. (Jack) Jiang. *Taming Non-Reproducibility: A Multi-Layered Framework for Divergence Investigation and Reproducible Execution in LLM-Powered Systems.* Under review, 2026.

– *B.Sc. Thesis:* Building a Recommendation System Including User-Specific Recommendation, Similar Items, and Cold Start. Amirkabir University of Technology, 2023. Released as the KabirRec Python library. GitHub PyPI

SKILLS

– **Languages**: Python, Java, C++, C, JavaScript

– **ML/LLM**: PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, vLLM, llama.cpp, RAGAS, OpenTelemetry

– **Infrastructure & Data**: Docker, Kubernetes, Linux, Git, CUDA, PostgreSQL, MongoDB, MySQL, Redis

– **English**: TOEFL iBT 117